

A Constraint-Based Probe for Exploring Phonological Sound-Space Without Decipherment

Abstract

This note describes an auxiliary, exploratory extension to a previously proposed phonological plausibility framework for evaluating unknown or ambiguous symbol strings. The extension introduces a phonological probe that generates broad sound-category constraints for structurally plausible symbol strings, while explicitly preventing phoneme assignment, lexical identification, or semantic interpretation.

The probe operates by conditioning sound-category generation on phonological profiles, enforcing conservative phonotactic filters, and applying entropy-based failure detection to prevent overfitting. The method is falsification-oriented: its primary function is to rule out unsafe or unjustified sound hypotheses rather than to propose readings. Failure cases are treated as informative outcomes.

This work is intended as a methodological demonstration for researchers in historical linguistics, undeciphered script studies, and comparative phonology, and makes no claims about any specific script or language. It extends the phonological plausibility framework described in *Basu, A. (2026). A Phonological Plausibility Framework for Evaluating Unknown or Ambiguous Symbol Strings*.

1. Background and motivation

Attempts to associate sounds with undeciphered symbol strings often fail due to premature commitment: symbol-to-sound mappings are proposed before structural compatibility is established. The phonological plausibility framework introduced earlier addresses this by separating orthographic parsing from phonological evaluation and by assessing strain across typologically motivated profiles.

However, even after plausibility filtering, users may be tempted to ask an unsafe question: “What might this sound like?”

The phonological probe described here exists to demonstrate how that question can be explored without answering it.

2. Design principles

The probe is governed by four principles:

1. **Structure precedes sound:** Only skeletons that survive plausibility filtering are considered.
 2. **Categories, not phonemes:** Sound values are represented as broad categories (for example, *stop*, *fricative*). They are not IPA segments or phonemic inventories.
 3. **Profile-conditioned constraints:** All sound-category constraints are relative to a phonological profile. The constraints are never absolute.
 4. **Failure is a valid outcome:** If constraints collapse or over-constrain the space, the probe stops.
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3. Method overview

Given an input symbol string:

1. Skeletons are generated via orthographic parsing.
2. Skeletons are evaluated against phonological profiles.
3. Profiles that can host a skeleton with low strain are retained.
4. For each retained profile:
 - Broad sound-category constraints are assigned for each position
 - Conservative phonotactic filters are applied
5. Remaining freedom in the sound-space is evaluated.
6. If entropy falls below a threshold, the probe fails.

At no point are real languages queried or modelled.

4. Entropy-based failure detection

Entropy is used here informally to describe degrees of freedom. Entropy is not for statistical inference. It is for ascertaining the degree of uncertainty that remains after the probe is applied. A high entropy indicates that almost anything is still possible, while a low entropy indicates that several possibilities have been eliminated. Entropy here is a diagnostic for collapse, not a measure of confidence.

A result is considered overdetermined and, therefore, unsafe if a process yields:

- Only one viable category for each position, or
- Only a negligible number of surviving combinations.

In such a case, the probe returns a failure state rather than a misleading output.

5. Interpretation of results

Successful probe output should be read as:

Assuming that the structure is real, in profile X, positions 1–n could plausibly belong to these broad sound classes.

Failure should be read as:

No sound-space exploration can be justified without introducing external assumptions.

Neither outcome supports decipherment.

6. Scope and non-goals

This method does not:

- Identify languages or families
- Reconstruct phonological systems
- Propose symbol-to-sound correspondences
- Validate readings of undeciphered scripts

It exists to constrain imagination, not to license it.

7. Conclusion

The phonological probe demonstrates that exploratory sound-space reasoning can be made safer by embedding it within a falsification-oriented framework. By placing failure over speculation, the method offers a way to think about sound without prematurely committing to it.

This work does not attempt decipherment. It explicitly prevents it by design, and any interpretation that treats its outputs as readings, reconstructions, or linguistic identifications is a category error.

8. References

Basu, A. (2026). A Phonological Plausibility Framework for Evaluating Unknown or Ambiguous Symbol Strings. Zenodo. <https://doi.org/10.5281/zenodo.18450309>